

Homework 2 - Potential Fields

Assigned - Nov 05, 2024, Due - Nov 19, 2024 (midnight)

For this homework, you are required to compare repulsive+attractive potential fields with navigation functions for a planar point robot in a 2D sphere world where both the obstacles and the boundary are circles. Your submission is expected to be a detailed report, describing the structure and design of your implementation, as well as various algorithmic components that you have implemented in the meantime. More importantly, your report should include results on how both algorithms work, provide both positive and negative example environments with different complexities. Your report should not include any code, except maybe small code snippets to support your descriptions if really necessary. Overall, your report should be structured like a good technical report or paper and be submitted as a PDF file together with your Matlab source files and any supporting media files such as videos. Please do not collaborate with your classmates except exchanging ideas and other inspirational materials.

The following steps can guide you through the implementation and the structure of your report.

- Implement a simple potential field planner with attractive and repulsive fields for the 2D sphere world with circular obstacles and boundary. The number, locations and sizes of obstacles should be configurable. Note that unlike the first homework, your planner has access to a global map for this homework. Investigate and report on the operation and performance of your planner implementation.
- Implement a navigation function based planner for the same domain, once again with configurable obstacles. Investigate and report on the operation and performance of your planner implementation.
- Construct a number of example settings with different number of obstacles and apply both algorithms to these domains. Compare the two implementation with respect to their performance, speed of convergence (i.e. how "long" does it take to get to the goal), difficulty of implementation and/or other issues you think are relevant. Feel free to propose/implement improvements on both algorithms based on your observations.

Once again, please make sure that your report is well organized into meaningful sections, with figures and captions properly references from within the text and an adequate number of citations to relevant resources from the literature. Note that using Matlab for visualization will probably a good idea. Another important issue is how you compute the path that results from following the negative gradient. I recommend that you use a continuous model and the ode45 function of Matlab to obtain your result accurately through numerical integration. However, you are allowed to use a discrete approximation with a properly chosen step size as well. Make sure to discuss related issues and potential problems in your report.

Submission

name and student ID. You should submit the following files on METU Class before the submission deadline: *lastname-firstname-report.pdf*, *lastname-firstname-sources.zip* with the ZIP file containing both the sources and the figures for your report as well as your implementation code and any other supporting material you might have. Your report should include your derivations, simulation results in the form of plots and detailed discussion on your findings and should be structured as a well-structured academic report or paper. **Late submissions will not be accepted.**